

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6563**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600024
		Course Section:	1		

L61_481_986_1_1505_48299



Name of the candidate. Atharva Vikas Khabale

Candidate's UID number:

2	0	2	2	6	0	0	0	2	4
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Examination class room number:

2	0	2
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Invigilator's signature with date:

 08/10/2025
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(To be filled in by the candidate)

EXAMINATION: MID SEM
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 08/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.1 a) $f(w) = \frac{1}{2} \|Xw - y\|^2$.

To prove: convergence requires learning rate $\eta < \frac{2}{\lambda_{\max}}$ where $\lambda_{\max}$ is largest value of eigenvalue of $X^T X$.

Let's first discuss what is gradient descent. It is a rule that explicitly helps in finding the optimal value of a parameter. It basically works on slope of intercept method, where the step value is calculated as.

$$\left(x_{k+1} = x_k - \eta \frac{df}{dx} \right)$$

$$\nabla f(x) = \frac{df(w)}{dx} = \frac{1}{2} \times 2 (xw - y) \times x$$

$$\nabla f(w) = (xw - y) \times w$$

$$\Rightarrow x_{k+1} = x_k - (x_k w - y) \times w$$

$$\left(\begin{aligned} x_{k+1} &= x_k - w^2 x_k + yw \\ x_{k+1} &= x_k (1 - w^2) + yw \end{aligned} \right)$$

$$\begin{aligned} x_{k+1} &= x_k - (x_k w - y) w \\ &= (x_k - w^2 x_k + yw) \end{aligned}$$

$$(x_{k+1} = (1 - w^2)x_k + yw)$$

$$\text{Hessian matrix} = \begin{vmatrix} \frac{d^2 f}{dx^2} & \frac{d^2 f}{dx dy} \\ \frac{d^2 f}{dy dx} & \frac{d^2 f}{dy^2} \end{vmatrix}$$

$$\frac{d^2 f}{dx^2} = \frac{d}{dx} (w(xw - y)) = w^2$$

$$\frac{d^2 f}{dy^2} = \frac{d}{dy} (x(xw - y) \times -1) = -1$$

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$$\frac{d^2 f}{dx dy} = \frac{d}{dx} ((xw-y)x-1) = -w.$$

$$\frac{d^2 f}{dy dx} = \frac{d}{dy} ((xw-y)xw) = -w$$

$$\Rightarrow \text{Hessian matrix } H = \begin{bmatrix} w^2 & -w \\ -w & 1 \end{bmatrix}$$

for eigen values $|H - \lambda I| = 0$.

$$\Rightarrow \begin{vmatrix} w^2 - \lambda & -w \\ -w & 1 - \lambda \end{vmatrix} = 0$$

$$\begin{aligned} \Rightarrow (w^2 - \lambda)(1 - \lambda) - w^2 &= 0 \\ \cancel{w^2} - \lambda - \lambda w^2 + \lambda^2 - \cancel{w^2} &= 0 \\ \lambda^2 - (1 + w^2)\lambda &= 0 \\ (\lambda = 0)(\lambda = 1 + w^2) \end{aligned}$$

for convergent to be maximum the two ends should be balanced & in opp direction & hence.

$$-(1 - \eta \lambda_{\min}) = 1 - \eta \lambda_{\max}$$

$$\begin{aligned} -1 + \eta \lambda_{\min} &= 1 - \eta \lambda_{\max} \\ \eta \lambda_{\min} + \eta \lambda_{\max} &= 2 \end{aligned}$$

$$\eta = \frac{2}{\lambda_{\min} + \lambda_{\max}}$$

either for max convergence $\eta = \frac{2}{\lambda_{\min} + \lambda_{\max}}$
for atleast convergence we consider $\lambda_{\min}$

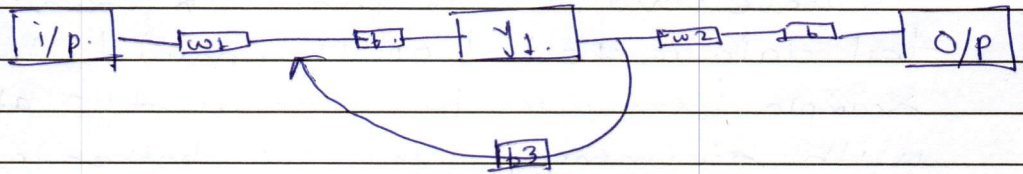
$$\Rightarrow \eta < \frac{2}{\lambda_{\max}}$$

[illegible]

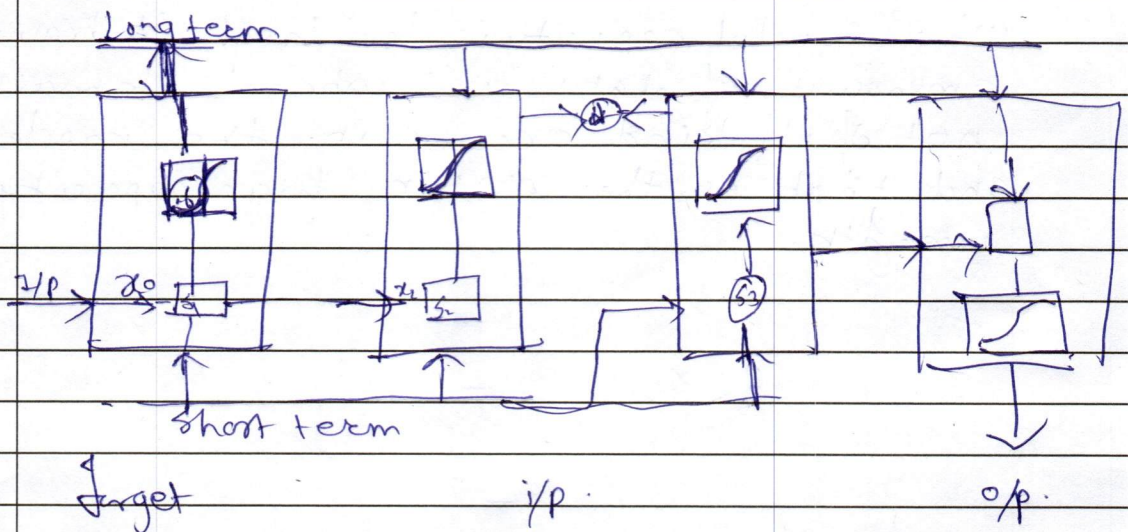
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Q.3. a. RNN.



LSTM.



So basically LSTM in contrast to RNN keeps two types of memories: one is long term and other is short term: while RNN reactively keeps track of its each and every previous O/p. LSTM simplifies it by converting it to two types of memory

① Long Term Memory

② Short-term memory

The memories depends on what is the imp of this o/p in further calculations. If it is going to be used in further calculations it will be stored in long term but if it is going to be required for a short duration it is stored in short-term memory.

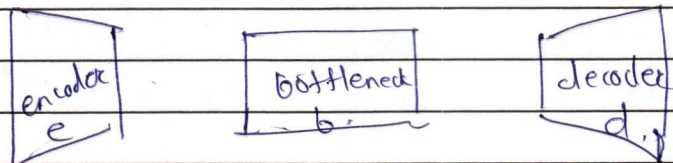


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Q-2 b. Under complete encoders have bottleneck.
i.e. the data at higher dimensions are projected to lower dimensions, we can say it as enhanced version of PCA.



$$\dim(e) > \dim(b) < \dim(d)$$

$\therefore$ the data from higher dimensions go in bottleneck & connect to lower dimension & then the same bottleneck project it or decode it to the o/p.

In over-complete autoencoders the scenario is quite different.

$$\dim(e) < \dim(b) > \dim(d)$$

i.e. dimensions of middle layer is higher than $\dim(e)$ and that is the reason why it learns identity function.

$\therefore$ The dimensions are remaining hence the remaining nodes just keeping quite start learning the identity funcn and it does not require regularisation cause nodes are already free and this chance for autoencoder is to learn how things & it learns the identity function.

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